

BLE Bootloader & DFU DEMO for nRF54l15

Introduction

Demo use SEGGER Embedded Studio Release 8.24 Build 2025041401.59002

nrfx 3.11 , nRF54L15 DK.

BLE_APP_54l15

BLE_APP_54l15 is a simple BLE app example with SoftDevice S115.

https://github.com/guangli2015/BLE_APP_nRF54L15.git

Memory setup is in Setup\nrf54l15_xxaa_application.ld

MEMORY

```
{  
    FLASH (rx) : ORIGIN = 0x0000b000, LENGTH = 0x146800  
    RAM (rwx) : ORIGIN = 0x20004400, LENGTH = 0x3B800  
}
```

bootloader_54l15

bootloader_54l15 is bootloader DEMO in 54l15 which perform Jumping to APP or entering DFU mode

https://github.com/guangli2015/bootloader_54l15.git

Memory setup is in Setup\nrf54l15_xxaa_application.ld

MEMORY

```
{  
    FLASH (rx) : ORIGIN = 0x00000000, LENGTH = 0x9000 /* Inside global RRAM0 */  
    RAM (rwx) : ORIGIN = 0x20004400, LENGTH = 0x3B800
```

}

Test step

1. Open BLE_APP_54l15.emProject and build entire solution
2. Open bootloader_54l15.emProject and build entire solution
3. nrfutil device recover
4. nrfutil device erase
5. cd C:\work\bootloader_54l15\Output\Debug\Exe
6. nrfjprog --program .\bootloader_54l15.hex --verify
7. cd C:\work\BLE_APP_54l15\Output\Debug\Exe
8. nrfjprog --program .\BLE_APP_54l15.hex --verify
9. cd C:\work\bootloader_54l15\Source\softdevice\s115
10. nrfutil device program --firmware s115_9.0.0-3.prototype_nrf54l15_softdevice.hex --options chip_erase_mode=ERASE_NONE

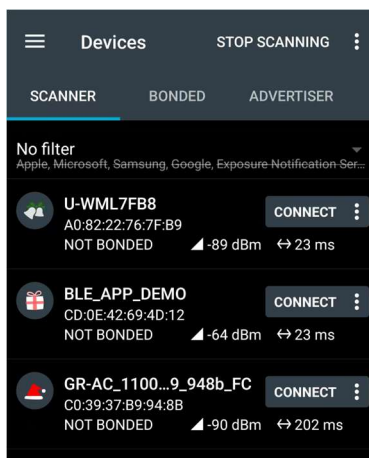
11. The debug log can be view using



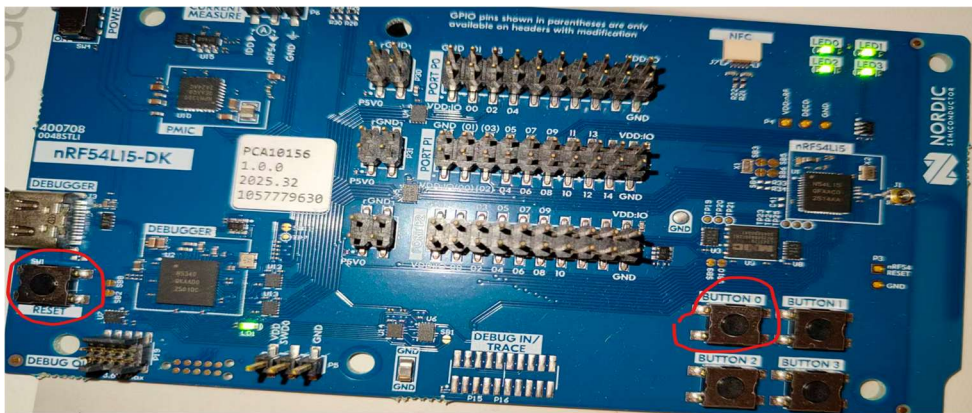
J-Link RTT Viewer V8.18
Compiled: 14:45:05 Mar 5 2025
<https://www.segger.com>



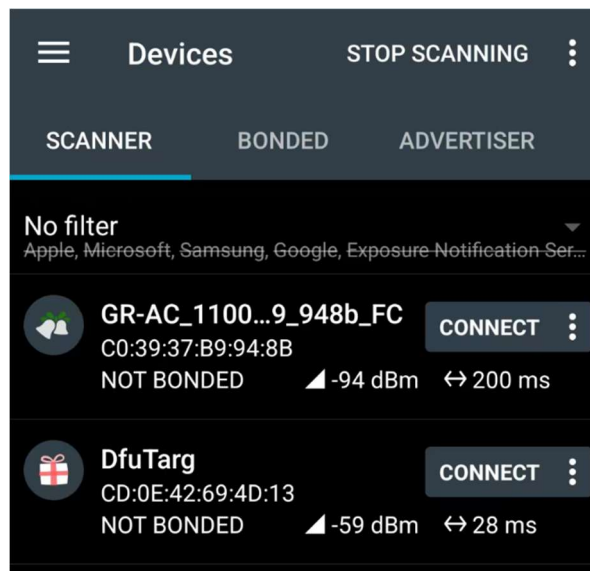
12. Open in your phone
13. Reset the board manually, Please check if have the adv device BLE_APP_DEMO



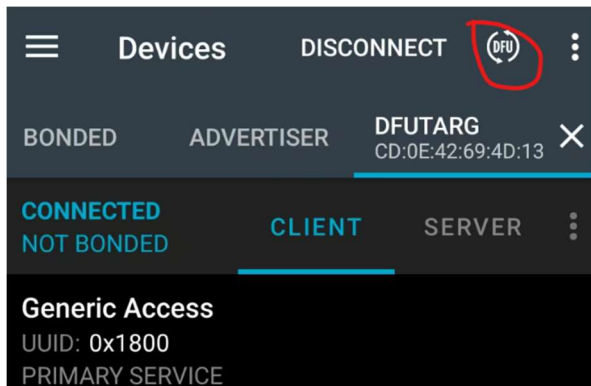
14. Open BLE_APP_54l15.emProject again and change Source\prj_config.h
CONFIG_BLE_ADV_NAME "BLE_APP_DEMO" to #define CONFIG_BLE_ADV_NAME
"BLE_APP_DEMO_new" and build
15. Cd C:\work\BLE_APP_54l15\Output\Debug\Exe & change 'BLE_APP_54l15.hex' into
'BLE_APP_54l15_new.hex'
16. nrfutil keys generate priv.pem
17. nrfutil nrf5sdk-tools pkg generate --application BLE_APP_54l15_new.hex --
application-version 2 --hw-version 52 --sd-req 0x100 --key-file priv.pem
BLE_APP_54l15_new.zip
18. transfer 'BLE_APP_54l15_new.zip' into your phone
19. To entering the DFU mode , Press the button 0 and reset at same time.



Then you can see the 'dfuTarg'



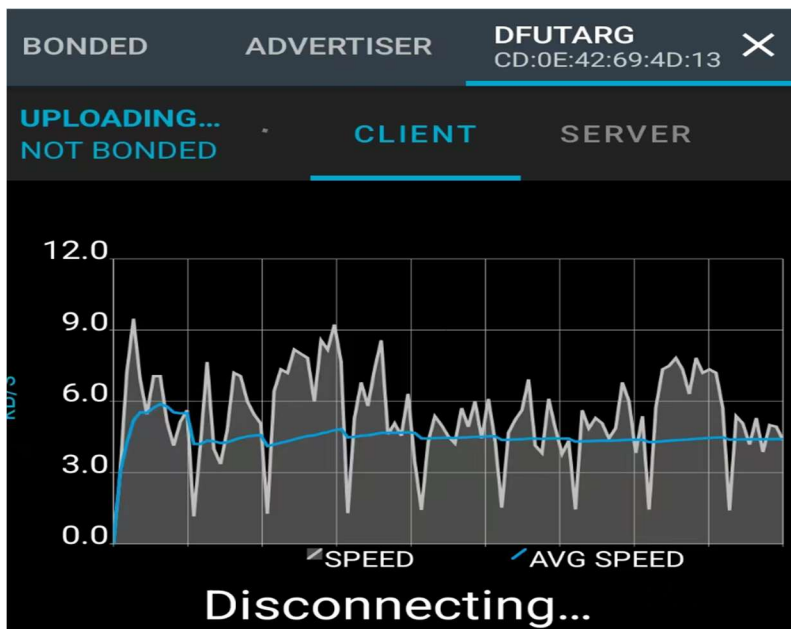
Connected and press DFU button in the up right corner



Select the 'BLE_APP_54l15_new.zip' in your phone



Reset your board manually after transfer finish



You will see the BLE_APP_DEMO_new

Devices

STOP SCANNING

SCANNER

BONDED

ADVERTISER

No filter

Apple, Microsoft, Samsung, Google, Exposure Notification Ser...



BLE_APP_DEMO_new

CD:0E:42:69:4D:12

NOT BONDED

CONNECT

-60 dBm

↔ 24 ms



U-WML7FB8

A0:82:22:76:7F:B9

NOT BONDED

CONNECT

-85 dBm

↔ 23 ms



GR-AC_1100...9_948b_FC

C0:39:37:B9:94:8B

NOT BONDED

CONNECT

-91 dBm

↔ 210 ms